# EXPERIMENT NO. 7

# CONFIGURATION OF DISTANCE VECTOR ROUTING PROTOCOL (RIP)

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## AIM

To study and configure RIP routing protocol and observe its routing behavior.

## OBJECTIVES

- To configure RIP protocol on multiple routers.
- To understand the distance vector routing mechanism.
- To observe routing table updates and hop count metrics.
- To analyze convergence behavior of RIP.
- To verify network connectivity using test packets.

## THEORY

Routing Information Protocol is a distance vector routing protocol that uses hop count as its metric. The maximum hop count supported by RIP is 15.

RIP version 2 supports classless routing, VLSM and multicast updates. It is suitable for small networks.

## SOFTWARE / MATERIALS REQUIRED

- Cisco Packet Tracer 8.x
- Computer System

## IMPORTANT COMMANDS / PROGRAM STEPS

```
router rip
version 2
network 192.168.1.0
network 10.0.0.0
show ip route
show ip protocols
ping 192.168.2.2
```

## PROCEDURE

1. Configure IP addresses on routers and PCs.
2. Enable RIP version 2 on all routers.
3. Advertise connected networks.
4. Verify routing table and RIP information.
5. Test connectivity between end devices.

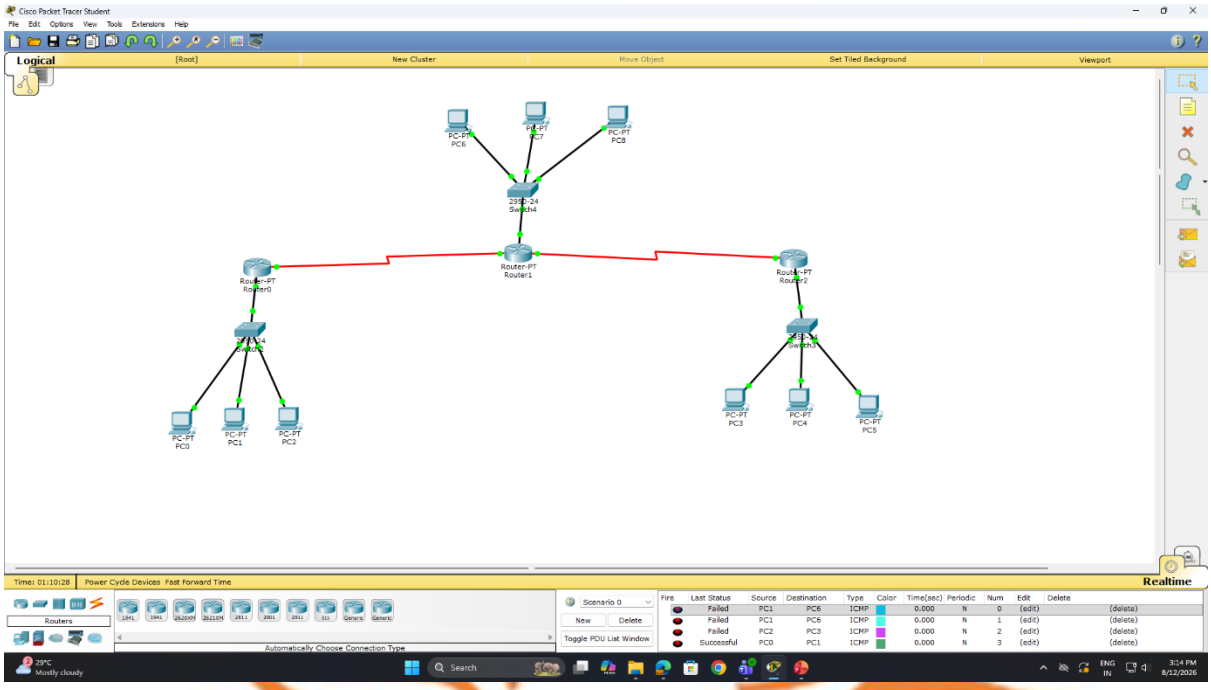
## OBSERVATION / COMPARISON TABLES

| Parameter | Observation |
|---|---|
| RIP Version | **Version 1 (RIPv1)** |
| Number of Routes Learned | **8** |
| Maximum Hop Count | **15 hops** |
| Routing Table Status | **RIP routes successfully learned and installed** |
| Ping Response | **Successful** |

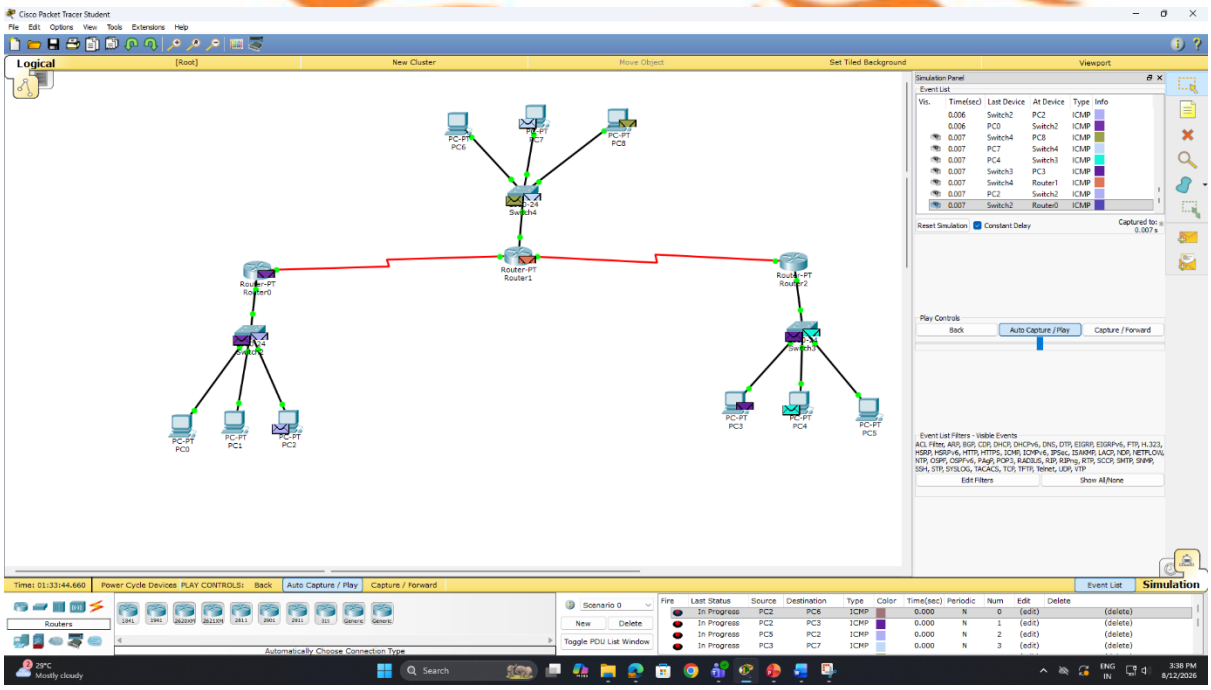
| Route Code | Description |
|---|---|
| R | RIP Route |
| C | Connected Route |
| S | Static Route |

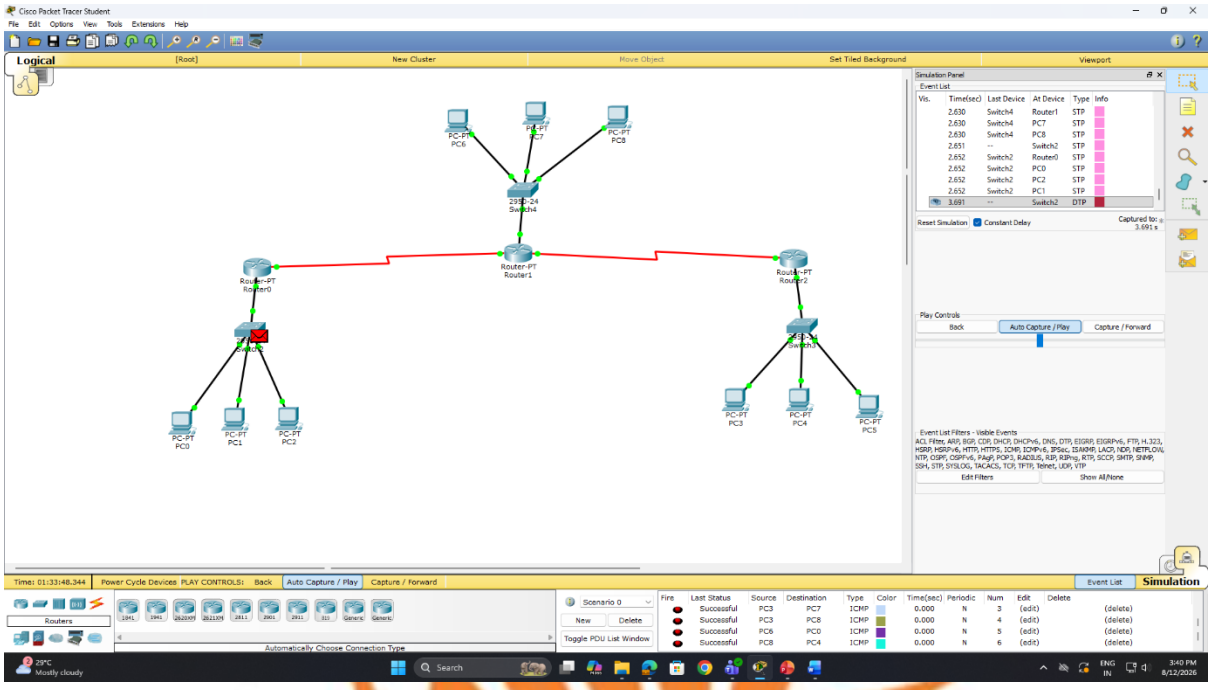
## RESULT:

## Overall:



## Scenario:




Cisco Packet Tracer Student
Logical
PC-PT PC6
PC-PT PC7
PC-PT PC8
2960-24 Switch4
Router-PT Router1
Router-PT Router0
Router-PT Router2
PC-PT PC0
PC-PT PC1
PC-PT PC2
PC-PT PC3
PC-PT PC4
PC-PT PC5
Simulation Panel
Event List
Play Controls
Back
Auto Capture / Play
Capture / Forward
Time: 01:33:48.344
Power Cycle Devices
PLAY CONTROLS:
Simulation
Routers
Automatically Choose Connection Type
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
Successful PC3 PC7 ICMP 0.000 N 3 (edit) (delete)
Successful PC3 PC8 ICMP 0.000 N 4 (edit) (delete)
Successful PC6 PC0 ICMP 0.000 N 5 (edit) (delete)
Successful PC8 PC4 ICMP 0.000 N 6 (edit) (delete)

## 1.1 Route Table before RIP of Router 1:

```
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, FastEthernet0/0
C    172.16.0.0/16 is directly connected, Serial2/0
Router#
```

## 1.2 Route Table After RIP of Router 1:

```
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, FastEthernet0/0
R    20.0.0.0/8 [120/1] via 172.16.0.2, 00:00:02, Serial2/0
R    30.0.0.0/8 [120/2] via 172.16.0.2, 00:00:02, Serial2/0
C    172.16.0.0/16 is directly connected, Serial2/0
R    192.168.0.0/24 [120/1] via 172.16.0.2, 00:00:02, Serial2/0
Router#
```

## 2.1 Route Table before RIP of Router 2:

```
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#ipishow ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    30.0.0.0/8 is directly connected, FastEthernet0/0
C    192.168.0.0/24 is directly connected, Serial2/0
```

## 2.2 Route Table After RIP of Router 2:

```
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

R    10.0.0.0/8 [120/2] via 192.168.0.1, 00:00:13, Serial2/0
R    20.0.0.0/8 [120/1] via 192.168.0.1, 00:00:13, Serial2/0
C    30.0.0.0/8 is directly connected, FastEthernet0/0
R    172.16.0.0/16 [120/1] via 192.168.0.1, 00:00:13, Serial2/0
C    192.168.0.0/24 is directly connected, Serial2/0
Router#
```

### 3.1 Route Table before RIP of Router 3:

```
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    20.0.0.0/8 is directly connected, FastEthernet0/0
C    172.16.0.0/16 is directly connected, Serial2/0
C    198.168.0.0/24 is directly connected, Serial3/0
```

### 3.2 Route Table After RIP of Router 3:

```
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

R    10.0.0.0/8 [120/1] via 172.16.0.1, 00:00:28, Serial2/0
C    20.0.0.0/8 is directly connected, FastEthernet0/0
R    30.0.0.0/8 [120/1] via 192.168.0.2, 00:00:17, Serial3/0
C    172.16.0.0/16 is directly connected, Serial2/0
C    192.168.0.0/24 is directly connected, Serial3/0
Router#
```

**CONCLUSION:**

Using the Cisco Packet Tracer CLI, we manually set up the RIP routing protocol on several routers. The devices automatically shared network topology details and updated their routing tables based on hop counts. Testing with ping utilities confirmed full end-to-end connectivity between different subnets, proving how Distance Vector Routing Protocols operate in practice.